# EXERCISE (O-1)

## SINGLE CORRECT TYPE QUESTIONS

### *General and Classification*

**1.** Drug which can block the binding site of the enzyme and prevent the binding of substrate are called _____.
(A) protein inhibitor (B) enzyme inhibitor (C) drug inhibitor (D) substrate inhibitor

**CEL011**

**2.** Drugs available for the treatment of a particular type of a problem is classified on the basis of ___.
(A) pharmacological effect (B) drug action
(C) chemical structure (D) molecular target

**CEL012**

**3.** Substrates bind to the site of the enzyme through interaction, like ___.
(A) ionic bonding, hydrogen bonding and disulphide interactions.
(B) dipole-dipole interaction, hydrogen bonding and disulphide interactions.
(C) hydrogen bonding, Van der Waals interaction and covalent bonding.
(D) ionic bonding, hydrogen bonding, Van der Waals interaction and dipole-dipole interactions.

**CEL013**

**4.** Which of the following statements is not true?
(A) Antihistamines interfere with the natural action of histamines by competing with histamines for binding sites of receptor.
(B) Antihistamine is a potent vasodilator.
(C) Histamines are responsible for allergic response to pollens.
(D) Histamine is a potent vasodilator.

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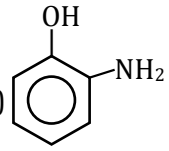
**CEL014**

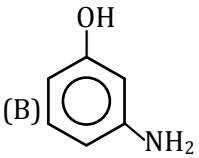
### *Drugs and Medicines*

**5.** The pair whose both species are used in antacid medicinal preparation is
(A) $NaHCO_3$ and $Mg(OH)_2$ (B) $Na_2CO_3$ and $Mg(OH)_2$
(C) $Ca(HCO_3)_2$ and $Mg(OH)_2$ (D) $Ca(OH)_2$ and $NaHCO_3$

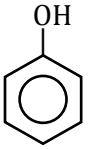
**CEL053**

**6.** Which of the following gives paracetamol on acetylation?

(A) 


(B) 


(C) 


(D) 


**CEL054**

**7.** Aspirin is called –
(A) Pyretic (B) Antiseptic (C) Antibiotic (D) Antipyretic

**CEL001**

**8.** An antipyretic is –
(A) Seldane (B) Paracetamol (C) Luminal (D) Aspartame

**CEL002**



For More Join: @IITJEE_Advanced

**9.** Medicine which is an antibiotic is –
(A) Ampicillin (B) Aspirin (C) Dimetapp (D) None of these
**CEL003**

**10.** Paracetamol is –
(A) Both antipyretic and analgesic (B) Analgesic
(C) Antipyretic (D) Antimalarial
**CEL004**

**11.** Sulpha drugs are derivatives of –
(A) Benzene sulphonic acid (B) Sulphanillic acid
(C) Sulphanilamide (D) p - aminobenzoic acid
**CEL005**

**12.** Which of the following is an antidiabetic drug –
(A) Insulin (B) Penicillin (C) Sucralose (D) Aspirin
**CEL006**

**13.** 2-Acetoxybenzoic acid is called –
(A) Antiseptic (B) Aspirin (C) Antibiotic (D) Mordant dye
**CEL007**

**14.** Arsenic drugs are mainly used in the treatment of–
(A) Jaundice (B) Typhoid (C) Syphilis (D) Cholera
**CEL008**

**15.** Aspirin is an acetylation product of –
(A) p-dihyroxybenzene (B) o-hydroxybenzoic acid
(C) o-dihydroxy benzene (D) m-hydroxybenzoic acid
**CEL009**

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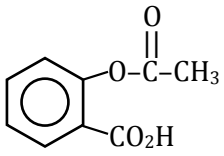
**16.** Chloramphenicol is an –
(A) Analgesic (B) Anaesthetic (C) Antibiotic (D) Antiseptic
**CEL010**

**17.** Which functional group is present in antacid cimetidine?
(A) Carboxylic acid (B) Amide (C) Alcohol (D) Cyanide
**CEL016**

**18.** Which of the following give violet-blue colour on reaction with neutral $FeCl_3$?
(A) Phenacitin (B) Morphine (C) Heroin (D) Chloramphenicol
**CEL017**

### *Uses of Different Types of Drugs and Medicines*

**19.** The following compound is used as




(A) an anti-inflammatory compound (B) analgesic
(C) hypnotic (D) antiseptic
**CEL051**



For More Join: @IITJEE_Advanced

**20.** Which of the following can possibly be used as analgesic without causing addiction and modification?
(A) Morphine (B) N-Acetyl-para-aminophenol
(C) Heroin (D) Codeine

**CEL052**

**21.** Drugs used to cure inflammation in the body are called ___.
(A) antiseptics (B) antibiotics (C) analgesics (D) antihistamines

**CEL015**

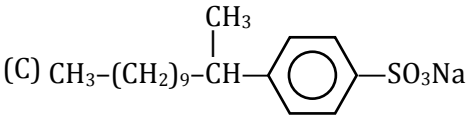
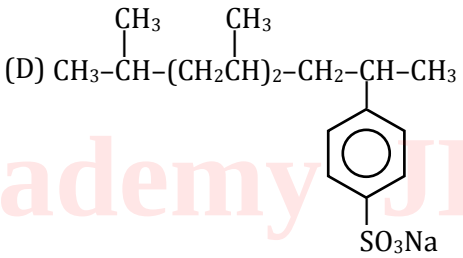
**22.** The drug given during hypertension is –
(A) Streptomycin (B) Chloroxylenol (C) Equanil (D) Aspirin

**CEL018**

### *Chemicals in Food; Soap and Detergents*

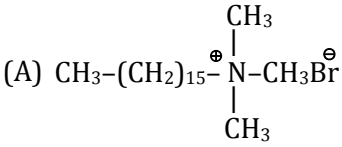
**23.** Choose the correct statement.
(A) Saccharin is 650 times sweeter than cane sugar.
(B) Alitame is 2000 times sweeter than cane sugar.
(C) Sucralose is 160 times sweeter than cane sugar.
(D) Aspartame is 550 times sweeter than cane sugar.

**CEL055**

**24.** Which of the following soap/detergents is the least biodegradable?
(A) $C_{17}H_{35}–COONa$ (B) $CH_3–(CH_2)_{11}–OSO_2Na$
(C) 
(D) 

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**CEL056**

**25.** Which one of the following is not a surfactant?
(A) 
(B) $CH_3 - (CH_2)_{14} – CH_2NH_2$
(C) $CH_3–(CH_2)_{16}\ CH_2OSO_2–Na^+$
(D) $OHC - (CH_2)_{14} - CH_2 - COO - Na^+$

**CEL057**

**26.** Which is correct reaction to prepare soap of glyceryl palmitate ?
(A) $(C_{15}H_{31}COO)_3C_3H_5 + 3NaOH \rightarrow (C_{15}H_{31}COONa)_3 + Glycerol$
(B) $(C_{17}H_{35}COO)_3C_3H_5 + 3NaOH \rightarrow (C_{17}H_{35}COONa)_3 + Glycerol$
(C) $(C_{17}H_{33}COO)_3C_3H_5 + 3NaOH \rightarrow (C_{17}H_{33}COONa)_3 + Glycerol$
(D) None of these

**CEL019**

**27.** Which of the following are anionic detergents?
(A) Sodium salts of sulphonated long chain alcohol.
(B) Ester of stearic acid and polyethylene glycol.
(C) Quaternary ammonium salt of amine with acetate ion.
(D) Product of stearic acid and polyethylene glycol.

**CEL020**



For More Join: @IITJEE_Advanced

# EXERCISE (O-2)

## MULTIPLE CORRECT TYPE QUESTIONS

### *General and Classification*

**1.** Which of the following statements are incorrect about receptor proteins?
(A) Majority of receptor proteins are embedded in the cell membranes.
(B) The active site of receptor proteins opens on the inside region of the cell.
(C) Chemical messengers are received at the binding sites of receptor proteins.
(D) Shape of receptor does not change during attachment of messenger.

**CEL059**

### *Uses of Different Types of Drugs and Medicines*

**2.** Compounds with antiseptic properties are
(A) $CHCl_3$ (B) $CHI_3$
(C) Boric acid (D) 0.3 ppm aqueous solution of $Cl_2$

**CEL060**

**3.** Which of the following statements are incorrect about penicillin ?
(A) An antibacterial drug (B) Ampicillium is its synthetic modification
(C) It has bacteriostatic effect (D) It is a broad spectrum antibiotic

**CEL061**

**4.** Amongst The following antihistamines, which are antacids?
(A) Ranitidine (B) Brompheniramine (C) Terfenadine (D) Cimetidine

**CEL062**

**5.** Which of the following are used for inducing sleep?
(A) Seconal (B) Serotonin (C) Luminal (D) Veronal

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**CEL063**

**6.** Broad spectrum antibiotics are
(A) penicillin (B) ofloxacin (C) aminoglycosides (D) chloramphenicol

**CEL064**

### *Chemicals in Food; Soap and Detergents*

**7.** Which of the following are not used as food preservatives ?
(A) Table salt (B) Sodium hydrogen carbonate
(C) Cane sugar (D) Benzoic acid

**CEL065**

## MATRIX MATCH TYPE QUESTION

### *Uses of Different Types of Drugs and Medicines*

**8.** Match the items given in column I with those in column II.

| Column I (Chemotherapeutic agents) | | Column II (Examples) | |
|---|---|---|---|
| 1. | Antibiotic | (a) | Chloroxylenol |
| 2. | Antiseptic | (b) | Chloropheniramine |
| 3. | Analgesic | (c) | Chloramphenicol |
| 4. | Tranquilizer | (d) | Diclophenac sodium |
| 5. | Antihistamine | (e) | Veronal |

(A) 1-c, 2-a, 3-d, 4-e, 5-b (B) 1-c, 2-a, 3-b, 4-e, 5-d
(C) 1-b, 2-e, 3-d, 4-a, 5-c (D) 1-c, 2-a, 3-d, 4-b, 5-e

**CEL058**



For More Join: @IITJEE_Advanced

# EXERCISE (S)

## Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

### *Drugs and Medicines*

**1.** Total number of tranquilizer drugs from the following is :
Veronal, Equanil, Iproniazid, Phenelzine, Aspirin, Ranitidine,
Pheniramine maleate, ofloxacin, Prontosil, Penicillin.

**CEL022**

**2.** Number of nitrogen atoms in one molecule of drug prontosil is____ .

**CEL024**

### *Chemicals in Food; Soap and Detergents*

**3.** Number of carbon atoms in one molecule of anionic detergent sodium laurylsulfate is ___.

**CEL030**

**4.** Number of chiral atoms in sucralose is:

**CEL021**

### *Uses of Different Types of Drugs and Medicines*

**5.** Number of $sp^2$ hybrid carbon atoms in one molecule of paracetamol is –

**CEL023**

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## Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

### *Drugs and Medicines*

**6.** Number of alcoholic group in codeine is ____.

**CEL029**

### *Uses of Different Types of Drugs and Medicines*

**7.** Degree of unsaturation of antibiotic penicillin is ____ .

**CEL025**



For More Join: @IITJEE_Advanced

# EXERCISE - JEE (Main) PYQ

**1.** The element playing significant role in neuromuscular function and interneuronal transmission is:
**[JEE (Main) 2023]**

(1) Be (2) Ca (3) Li (4) Mg

**CEL069**

**2.** Some drugs bind to a site other than, the active site of an enzyme. This site is known as
**[JEE (Main) 2022]**

(1) non-active site (2) allosteric site (3) competitive site (4) therapeutic site

**CEL041**

**3.** Given below are two statements : One is labelled as **Assertion A** and the other labelled as **Reason R**.
**Assertion A:** Antihistamines do not affect the secretion of acid in stomach.
**Reason R :** Antiallergic and antacid drugs work on different receptors.
In the light of the above statements, choose the **correct** answer from the options given below:
**[JEE (Main) 2023]**

(1) A is false but R is true
(2) Both A and R are true and R is the correct explanation of A
(3) A is true but R is false
(4) Both A and R are true but R is not the correct explanation of A.

**CEL045**

**4.** Amongst the following compounds, which one is an antacid? **[JEE (Main) 2023]**

(1) Ranitidine (2) Meprobamate (3) Terfenadine (4) Brompheniramine

**CEL046**

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**5.** Among the following, the number of tranquilizer/s is/are_____. **[JEE (Main) 2023]**

(1) Chloroliazepoxide (2) Veronal (3) Valium (4) Salvarsan

**CEL049**

**6.** Given below are two statements : **[JEE (Main) 2023]**
**Statement I :** Noradrenaline is a neurotransmitter.
**Statement II :** Low level of noradrenaline is not the cause of depression in human.
In the light of the above statements, choose the correct answer from the options given below
(1) Statement I is correct but Statement II is incorrect
(2) Statement I is incorrect but Statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

**CEL066**

**7.** Which of the following is not a broad spectrum antibiotic? **[JEE (Main) 2022]**

(1) Vancomycin (2) Ampicillin (3) Ofloxacin (4) Penicillin G

**CEL039**

**8.** Which of the following statements is incorrect for antibiotics? **[JEE (Main) 2023]**
(1) An antibiotic must be a product of metabolism.
(2) An antibiotic is a synthetic substance produced as a structural analogue of naturally occurring antibiotic.
(3) An antibiotic should promote the growth or survival of microorganisms.
(4) An antibiotic should be effective in low concentrations.

**CEL043**



For More Join: @IITJEE_Advanced

**9.** A doctor prescribed the drug Equanil to a patient. The patient was likely to have symptoms of which disease? **[JEE (Main) 2023]**

(1) Stomach ulcers
(2) Hyperacidity
(3) Anxiety and stress
(4) Depression and hypertension

**CEL044**

**10.** Match List-I with List-II

| | List-I | | List-II |
|---|---|---|---|
| (A) | Tranquilizers | (I) | Anti blood clotting |
| (B) | Aspirin | (II) | Salvarsan |
| (C) | Antibiotic | (III) | Antidepressant drugs |
| (D) | Antiseptic | (IV) | Soframicine |

Choose the correct answer from the options given below: **[JEE (Main) 2023]**

(1) (A) – IV, (B) – II, (C) – I, (D) – III
(2) (A) – II, (B) – I, (C) – III, (D) – IV
(3) (A) – III, (B) – I, (C) – II, (D) – IV
(4) (A) – II, (B) – IV, (C) – I, (D) – III

**CEL050**

**11.** Match List I with List II **[JEE (Main) 2023]**

| | List-I Type | | List-II Name |
|---|---|---|---|
| (A) | Antifertility drug | (I) | Norethindrone |
| (B) | Tranquilizer | (II) | Meprobomate |
| (C) | Antihistamine | (III) | Seldane |
| (D) | Antibiotic | (IV) | Ampicillin |

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Choose the correct answer from the options given below:

(1) A-II, B-I, C-III, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-I, B-II, C-III, D-IV

**CEL067**

**12.** Given below are two statements :- **[JEE (Main) 2023]**

**Statement-**I : Morphine is a narcotic analgesis. It helps in relieving pain without producing sleep.

**Statement-**II : Morphine and its derivatives are obtained from opium poppy.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Statement I is true but Statement II is false
(2) Both Statement I and Statement II are false
(3) Both Statement I and Statement II are true
(4) Statement I is false but Statement II is true

**CEL071**

**13.** The statement/s which are true about antagonists from the following is/are :- **[JEE (Main) 2023]**

A. They bind to the receptor site.
B. Get transferred inside the cell for their action.
C. Inhibit the natural communication of the body.
D. Mimic the natural messenger.

Choose the correct answer from the options given below :-

(1) B only (2) A, C and D (3) A and B (4) A and C

**CEL073**



For More Join: @IITJEE_Advanced

**14.** Drugs used to bind to receptors, inhibiting its natural function and blocking a message are called:
**[JEE (Main) 2022]**

(1) Agonists (2) Antagonists (3) Allosterists (4) Anti histaminists

**CEL040**

**15.** Given below are two statements, one is labelled as **Assertion A** and the other is labelled as **Reason R**
**Assertion A :-** Butylated hydroxyl anisole when added to butter increases its shelf life.
**Reason R :-** Butylated hydroxyl anisole is more reactive towards oxygen than food.
In the light of the above statements, choose the most appropriate answer from the options given below:-
**[JEE (Main) 2023]**

(1) Both A and R are correct and R is the correct explanation of A.
(2) A is correct but R is not correct.
(3) A is not correct but R is correct.
(4) Both A and R are correct but R is NOT the correct explanation of A.

**CEL042**

**16.** Match List I with List II **[JEE (Main) 2023]**

| **List-I** | | **List-II** | |
|---|---|---|---|
| **Antimicrobials** | | **Name** | |
| (A) | Narrow Spectrum Antibiotic | (I) | Furacin |
| (B) | Antiseptic | (II) | Sulphur dioxide |
| (C) | Disinfectants | (III) | Penicillin-G |
| (D) | Broad spectrum antibiotic | (IV) | Chloramphenicol |

(1) (A)–III, (B)–I, (C)–II, (D)-IV (2) (A)–I, (B)–II, (C)–IV, (D)-III
(3) (A)–II, (B)–I, (C)–IV, (D)-III (4) (A)–III, (B)–I, (C)–IV, (D)-II

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**CEL068**

**17.** Match List I with List II **[JEE (Main) 2023]**

| List I | | List II | |
|---|---|---|---|
| A. | Saccharin | I. | High potency sweetener |
| B. | Aspartame | II. | First artificial sweetening agent |
| C. | Alitame | III. | Stable at cooking temperature |
| D. | Sucralose | IV | Unstable at cooking temperature |

Choose the correct answer from the options given below:
(1) A-II, B-III, C-IV, D-I (2) A-II, B-IV, C-III, D-I
(3) A-IV, B-III, C-I, D-II (4) A-II, B-IV, C-I, D-III

**CEL072**

**18.** The reaction used for preparation of soap from fat is: **[JEE (Main) 2023]**

(1) reduction reaction (2) alkaline hydrolysis reaction
(3) an addition reaction (4) an oxidation reaction

**CEL074**



For More Join: @IITJEE_Advanced

# ANSWER KEY

## EXERCISE (O-1)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | A | D | B | A | C | D | B | A | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | C | A | B | C | B | C | D | B | B | B |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | | | |
| Ans. | D | C | B | D | B | A | A | | | |

## EXERCISE (O-2)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Ans. | BD | BC | CD | AD | ACD | BCD | BD | A |

## EXERCISE (S)

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Ans. | 4 | 5 | 12 | 9 | 7 | 1.00 | 5.00 |

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## EXERCISE - JEE (Main) PYQ

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 2 | 2 | 2 | 1 | 3 | 1 | 4 | 3 | 4 | 3 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | | |
| Ans. | 4 | 4 | 4 | 2 | 1 | 1 | 4 | 2 | | |



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